

Dr. Sanoj Kumar

Sr. Associate Professor, Data Science Cluster, SOCS, UPES, Dehradun, Uttarakhand, India.

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Office Address:

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Permanent Address:

Dr. Sanoj Kumar S/o Shri. Pal Singh Saini
Village + Post : Manakpur Adampur
District: Haridwar
Pin: 247668, Uttarakhand, India

Professional Experience:

March 2026 – Till date	Sr. Associate Professor <i>Institute:</i> Data Science Cluster, SOCS, UPES, Dehradun, Uttarakhand, India. <i>Duties:</i> Assigned by the Department including lectures, tutorials and research work.
March 2025 – Till date	Associate Professor <i>Institute:</i> Data Science Cluster, SOCS, UPES, Dehradun, Uttarakhand, India. <i>Duties:</i> Assigned by the Department including lectures, tutorials and research work.
June 2023 – Till date	Assistant Professor (SG) <i>Institute:</i> Data Science Cluster, SOCS, UPES, Dehradun, Uttarakhand, India. <i>Duties:</i> Assigned by the Department including lectures, tutorials and research work.
January 2018 – May 2023	Assistant Professor (SG) <i>Institute:</i> Department of Mathematics, University of Petroleum and Energy Studies, Dehradun, Dehradun, Uttarakhand, India. <i>Duties:</i> Assigned by the Department including lectures, tutorials and research work.
January 2015 – December 2017	Assistant Professor (SS) <i>Institute:</i> Department of Mathematics, University of Petroleum and Energy Studies, Dehradun, Dehradun, Uttarakhand, India. <i>Duties:</i> Assigned by the Department including lectures, tutorials and research work.
October 2013 – September 2014	Postdoctoral Fellow <i>Institute:</i> Department of Mathematics and Computer Science, University of Udine, Udine, Italy.
2008-2013	Teaching Assistant <i>Institute:</i> Department of Mathematics, Indian Institute of Technology Roorkee, India. <i>Duties:</i> Conducting tutorials and labs for graduate and under-graduate students for Calculus, Numerical Analysis, Image Processing, and Computer Applications.
2008-2013	MHRD Research Fellow <i>Institute:</i> Department of Mathematics, Indian Institute of Technology Roorkee, India.
2006-2008	Guest Lecturer <i>Institute:</i> Department of Mathematics, Motilal Nehru National Institute of Technology Allahabad, India. <i>Duties:</i> Conducting lectures, tutorials and labs for graduate and under-graduate students for Mathematics (I, II, III), Numerical Analysis and Business Mathematics.

Education:

2008-2013	Ph.D. (Applied Mathematics) <i>Institute:</i> Department of Mathematics, Indian Institute of Technology Roorkee, India. <i>Thesis Title:</i> Robust Estimation of Optical flow and Disparity Map from Image Sequences. <i>Advisors:</i> Prof. N. Sukavanam and Dr. R. Balasubramanian. <i>Date of Registration:</i> 02/08/2008, <i>Date of Submission:</i> 27/12/2012 <i>Date of viva-voce:</i> 24/05/2013, <i>Date of Award:</i> 21/09/2013
2003-2005	M.Sc. (Mathematics) <i>University:</i> Ch. Charan Singh University, Meerut, India. <i>Division - First.</i>
2000-2003	B.Sc. (Mathematics) <i>University:</i> Ch. Charan Singh University, Meerut, India. <i>Division - First.</i>
2000	Intermediate <i>College:</i> Ch. Bharat Singh DAV Inter College, Jhabrera, Haridwar, Uttarakhand, India.

Thesis Supervised:

PhD	04 ongoing
M.Tech.	01 Completed
MCA (Minor project)	03 Completed
B.Tech. (Minor project)	16 Completed
B.Sc. (Dissertation)	02 Completed
BCA	02 Completed

Projects:

S.N.	Title of Project	Sponsering Agency	Cost of Project	Role	Status
1	Deep Learning: From Foundations to Cutting- Edge Technique	SERB	INR 2,00,000.00	Convenors	Aawrded
2	AI-Powered Climate-Smart Agriculture for Rural India and Italy	Indoitalian Executive Programme Of Cooperation In Scientific & Technological Cooperation (Exchange of researchers)	INR 11,97,000.00	Co-Principal Investigator	Submitted

Research Interests:

Mathematics

Mathematical Statistics, Numerical Analysis, Optimization Techniques

Computer Science and Engineering

Digital Image Processing, Computer Vision, Machine learning, and Deep learning

Publications: The details of my publications are listed as below:**Journals:**

1. Rahul Pal, **Sanoj Kumar** and Gaurav Bhatnagar, "Brain MRI Segmentation using Deep Learning: A Review", **under review** in *Neurocomputing* (Elsevier), 2026. (SCI: IF 6.5)
2. Rahul Pal, **Sanoj Kumar**, Gaurav Bhatnagar, "Uncertainty-aware multi-class brain tumor segmentation using Bayesian U-Net variants", **published** in *Biomedical Physics & Engineering Express*, 12(2), 025076, 2026. (SCI: IF- 2.0)
3. Swati Rastogi, **Sanoj Kumar**, Musrrat Ali, Abdul Rahaman Wahab Sait, "Sequential Multimodal Biometric Authentication Fusion System", **published** in *Mathematics*, 14(7), 1178, 2026. (SCI: IF- 2.3)
4. **Sanoj Kumar**, Harendra Pal Singh, Musrrat Ali, Abdul Rahaman Wahab Sait, "Earthquake-Resilient Structural Control Using PSO-Based Fractional Order Controllers", **published** in *Fractal and Fractional*, 9(12), 759, 2025. (SCI: IF- 3.5)
5. Musrrat Ali, Deepika Saini, **Sanoj Kumar**, Abdul Rahaman Wahab Sait, "Optimization-Driven Reconstruction of 3D Space Curves from Two Views Using NURBS", **published** in *Mathematics*, 13(14), 2256, 2025. (SCI: IF- 2.3)
6. Manoj Kumar Singh, **Sanoj Kumar**, and Deepika Saini, "A Complex Network Analysis of Image Watermarking Scheme Based on SVD and DWT" **published** in *SN Computer Science*, 5(8), 1009, 2024.

7. Himanshu Prasad Mangain, Rahul Pal, **Sanoj Kumar**, Ranjeet Brajpuriya, Jitendra K Pandey, "Machine Learning-Based Stability Prediction and Analysis of Polypropylene Cu-MA Superhydrophobic Coating on the Aluminum Substrate", **published** in The Journal of Physical Chemistry C, 35, 2011-2022, 2024. DOI 10.1016/j.jmrt.2025.01.127, (SCI: IF- 3.3)
8. Manoj K. Singh, Deepika Saini, and **Sanoj Kumar**, "Topological analysis of image reconstruction based on polar complex exponential transform", **published** in Journal of Computational and Cognitive Engineering, 2024.
9. Shubh Gupta, **Sanoj Kumar**, Karan Singh, Deepika Saini, "Unveiling anomalies: harnessing machine learning for detection and insights", **published** in Engineering Research Express, 6(3), 5215, 2024. DOI 10.1088/2631-8695/ad66b2. (SCI: IF- 1.5)
10. Musrrat Ali, Deepika Saini, **Sanoj Kumar**, "Inverse Geometric Reconstruction Based on MW-NURBS Curves", published in Mathematics, 12 (13), 2071, 2024. (SCI: IF- 2.2)
11. Rahul C. Pal, **Sanoj Kumar**, Manoj K. Singh, "Topological Data Analysis and Image Visibility Graph for Texture Classification", **published** in *International Journal of System Assurance Engineering and Management*, Springer, 2024, pp 1-11. (SCI: IF- 2.018)
12. Musrrat Ali, Lakshay Goyal, CM Sharma, and Sanoj Kumar, "Edge Computing Enabled Abnormal Activity Recognition for Visual Surveillance", **published** in *Electronics* 2024, 13(2), 251. <https://doi.org/10.3390/electronics13020251>, (SCI: IF- 2.690)
13. Musrrat Ali, and Sanoj Kumar, "A Robust Zero-Watermarking Scheme in Spatial Domain by Achieving Features Similar to Frequency Domain" **published** in *Electronics* 2024, 13(2), 435. <https://doi.org/10.3390/electronics13020435>, (SCI: IF- 2.690)
14. Musrrat Ali, **Sanoj Kumar**, Rahul Pal, Manoj K Singh, Deepika Saini, "Graph-and Machine-Learning-Based Texture Classification", **published** in *Electronics* 2023, 12(22), 4626. <https://doi.org/10.3390/electronics12224626>, (SCI: IF- 2.690)
15. Pooja Singh, Neeru Rathee, Sunanda Sharda, Sanoj Kumar, "Comparative Study of Rough Set-Based FCM and K-Means Clustering for Tumor Segmentation from Brain MRI Images", **published** in *Revue d'Intelligence Artificielle* Vol. 37, No. 4, August, 2023, pp. 921-927.
16. Manoj K Singh, Sanoj Kumar, Gaurav Bhatnagar, Deepika Saini, Musrrat Ali, Chandra Mani Sharma, Navel Sharma, "A Blend of Analytical and Numerical Methods to Compute Orthogonal Image Moments over a Unit Disk" **published** in *Wireless Communications and Mobile Computing*, vol. 2022, Article ID 1344584, 15 pages, 2022. <https://doi.org/10.1155/2022/1344584>. (SCI: IF - 2.146)
17. **Sanoj Kumar**, Gaurav Bhatnagar, Girish Dobhal, Manoj K. Singh and Deepika Saini, "A secure and robust stereo image encryption algorithm based on DCT and Schur decomposition", **published** in *Journal of Information Technology Management*, 14, pp 23 – 43, 2022. (Scopus)
18. Manoj K. Singh, **Sanoj Kumar**, Musrrat Ali and Deepika Saini, "Application of a novel image moment computation in X-ray and MRI image watermarking" **published** in *IET Image Processing*, 11, pp 666 – 682, 2021. (SCI: IF - 2.373)
19. Deepika Saini, **Sanoj Kumar**, Manoj K. Singh and Musrrat Ali, "Two View NURBS reconstruction based on GACO model" **published** in *Complex & Intelligent Systems* (Springer) 7 (5), 2329-2346, 2021. (SCI: IF – 6.700)

20. **Sanoj Kumar**, Manoj K. Singh and Deepika Saini, “An image watermarking framework based on PSO and FrQWT”, **published** in *Journal of Discrete Mathematical Sciences and Cryptography* (Taylor & Francis) 24 (5), 1293-1308, 2021. (SCI: IF – 0.68)
21. Musrrat Ali, Chang Wook Ahn, Millie Pant, **Sanoj Kumar**, Manoj K. Singh and Deepika Saini, “An Optimized Digital Watermarking Scheme Based on Invariant DC Coefficients in Spatial Domain”, **published** in *Electronics* 9(9), 1428 (1-17), [10.3390/electronics9091428](https://doi.org/10.3390/electronics9091428) , 2020. (SCI: IF- 2.690)
23. **Sanoj Kumar**, Sanjeev Kumar, Balasubramanian Raman and N. Sukavanam, “Dual Tree Fractional Quaternion Wavelet Transform for Disparity Estimation”, **published** in *ISA Transactions (Elsevier)* 53(2), pp 547-559, 2014. (SCI: IF – 5.991)
24. **Sanoj Kumar**, Sanjeev Kumar, Nagarajan Sukavanam and Balasubramanian Raman, “Image Disparity Estimation using Fractional Dual-Tree Complex Wavelet Transform: A Multi-Scale Approach”, **published** in *International Journal of Wavelets, Multiresolution and Information Processing (IJWMIP)* 11, 1350004 (2013) [21 pages] DOI: 10.1142/S0219691313500045. (SCI: IF – 1.408)
25. **Sanoj Kumar**, Sanjeev Kumar, Nagarajan Sukavanam and Balasubramanian Raman, “Human Visual System and Segment-Based Disparity Estimation”, **published** in *International Journal of Electronics and Communications (Elsevier)* 67(5), pp 372-381, 2013. (SCI: IF – 3.183)
26. **Sanoj Kumar**, Gaurav Bhatnagar, Balasubramanian Raman and N. Sukavanam, “Security of Stereo Images during Communication and Transmission”, **published** in *Advanced Science Letters* 6, pp 173—179, 2012.

Conferences :

1. Rahul Pal, Sanoj Kumar, Gaurav Bhatnagar, “Uncertainty-Aware Brain Tumor Segmentation”, published in 2025 IEEE International Conference on Computer Vision and Machine Intelligence (CVMI), 12-13 October 2025, Department of Electronics and Communication Engineering National Institute of Technology (NIT), Rourkela. Rourkela, Odisha, India – 769008.
2. Rahul C. Pal, **Sanoj Kumar**, Manoj K. Singh, “Integrating Image Visibility Graph and Topological Data Analysis for Enhanced Texture Classification”, **published** in International conference on soft computing for problem-solving, Springer Nature Singapore, 2024/07/01.
3. Manoj K. Singh, **Sanoj Kumar**, Deepika Saini and Gaurav Bhatnagar, “A reversible and rotational invariant watermarking scheme using polar harmonic transforms”, **published** in Academia-Industry Consortium for Data Science (AICDS-2020).
4. **Sanoj Kumar**, Gaurav Bhatnagar, Girish Dobhal, Manoj K. Singh and Deepika Saini, “A secure and robust stereo image encryption algorithm based on DCT and Schur decomposition”, **presented** in International Conference on Communication and Computing Systems (ICCCS - 2021), February 5-7, 2021, ICARS, New Delhi.
5. **Sanoj Kumar**, Manoj K. Singh and Deepika Saini, “An image watermarking framework based on PSO and FrQWT”, **presented** in 2nd International Conference on Networks and Cryptology (NETCRYPT - 2020), December 4-6, 2020, New Delhi.
6. **Sanoj Kumar**, Manoj K. Singh, Musrrat Ali and Deepika Saini, “A robust medical image watermarking framework based on SVD and DE in Integer DCT domain ”, **published** in 2020 IEEE Sixth International Conference on Multimedia Big Data (BigMM), September 24-26, 2020, New Delhi.

7. **Sanoj Kumar**, Claudio Picciarelli and Harendra Pal Singh, “Reconfiguration of PTZ Camera Network with minimum resolution”, **published** in 4th International Conference on Harmony search, Soft computing and Applications (ICHSA 2018), BML Munjal University, Gurgaon, India.
8. **Sanoj Kumar**, Sanjeev Kumar, and Anuj Kumar, “Histogram based Motion Estimation of Underwater Images”, **published** in International Conference on Frontiers in Industrial and Applied Mathematics (FIAM 2018), NIT Hamirpur, Hamirpur, Himachal Pradesh, India.
9. **Sanoj Kumar**, Sanjeev Kumar, and Balasubramanian Raman, “A Variational Approach for Optical Flow Estimation in Infra-Red or Thermal Images”, **published** in Second International Conference on Image Information Processing (ICIIP -2013) December 9-11, 2013, Jaypee University of Information Technology Wanknaghat, Shimla, Himachal Pradesh, India.
10. **Sanoj Kumar**, Sanjeev Kumar, Nagarajan Sukavanam and Balasubramanian Raman, “Optical flow Estimation using Fractional Quaternion Wavelet Transform”, **published** in International Conference on Industrial and Intelligent Information (ICII -2012) March 17-18, 2012, Singapore.
11. **Sanoj Kumar**, Nagarajan Sukavanam, Balasubramanian Raman and Sanjeev Kumar, “Human Visual System and Wavelet transform based disparity estimation”, **published** in Fourth International Conference on Emerging Trends in Engineering and Technology (ICETET), November 18-20, 2011, Mauritius.
12. **Sanoj Kumar**, Sanjeev Kumar, Nagarajan Sukavanam and Balasubramanian Raman, “Disparity Estimation using Fractional Dual Tree Complex Wavelet Transform”, **published** in International Conference on Image Information Processing (ICIIP -2011) November 3-5, 2011, Jaypee University of Information Technology Wanknaghat, Shimla, Himachal Pradesh, India.
13. **Sanoj Kumar**, Sanjeev Kumar, Balasubramanian Raman and Nagarajan Sukavanam, “Human Action Recognition in a Wide and Complex Environment”, **published** in Real-Time Image and Video Processing 2011, part of IS&T/SPIE Electronic Imaging 2011, Jan 23-27 2011, San Francisco, California, USA.

Bookchapters:

1. Mukesh Kumar Awasthi, **Sanoj Kumar**, Deepika Saini, “Introduction to AI-Driven Mathematical Modeling”, **published** in Artificial Intelligence Techniques in Mathematical Modeling and Optimization, CRC Press, Taylor and Francis, Pages 1-12, 2026.
2. Manoj Kumar Singh, **Sanoj Kumar**, Deepika Saini, “Deep Learning Techniques From Training to Generalization”, **published** in Artificial Intelligence Techniques in Mathematical Modeling and Optimization, CRC Press, Taylor and Francis, Pages 57-66, 2026.
3. **Sanoj Kumar**, Mukesh Kumar Awasthi, Deepika Saini, “Future Trends and Emerging Technologies in AI Optimization”, **published** in Artificial Intelligence Techniques in Mathematical Modeling and Optimization, CRC Press, Taylor and Francis, Pages 308-324, 2026.
4. **Sanoj Kumar**, Zahid Akhtar, Harsh Satsangi, Sakshi Sehrawat, Namit Arora, Kartik Bamal, “Depression Prediction Using Machine Learning Techniques” published in Artificial Intelligence in Healthcare, CRC Press, Taylor and Francis, 2024.
5. **Sanoj Kumar**, Rahul Pal, Niki Martinel, Deepika Saini, “AI Chatbot for Depression Self-Help”, published in Artificial Intelligence in Healthcare, CRC Press, Taylor and Francis, 2024.
6. **Sanoj Kumar**, Manoj K. Singh and Deepika Saini, “Image watermarking with polar harmonic moments” published in Computing and Simulation for Engineerers published by CRC Press, Taylor and Francis, 2021.

7. **Sanoj Kumar**, Gaurav Bhatnagar, "SIE: An application to secure Stereo Images using Encryption", published in Handbook of Multimedia Information Security: Techniques and Applications (HMIS), Springer, October 2018.

Book:

1. Gaurav Bathla, **Sanoj Kumar**, Harish Garg and Deepika Saini, "Artificial Intelligence in Healthcare: Emphasis on Diabetes, Hypertension and Depression management", edited book **published** to the book series *Intelligent Data-Driven Systems and Artificial Intelligence*, CRC Press, 2024.
2. Mukesh Kumar Awsthi, **Sanoj Kumar** and Deepika Saini, "Artificial Intelligence Techniques in Mathematical Modeling and Optimization", edited book **published** to the book series *Intelligent Data-Driven Systems and Artificial Intelligence*, CRC Press, 2025.

Attended Conferences/Workshop/Seminar/Symposium/Short Term Course:

1. International conference on intelligent computing and communication techniques at School of Computer and Systems Sciences, JNU, New Delhi, India from June 28-29, 2024.
2. International Congress and 8th conference of Indian Society of Industrial & Applied Mathematics on certain emerging areas in applicable Mathematics at Department of Mathematics, University of Jammu, Jammu, India from 31st March - 3rd April, 2007.
3. A workshop on Year long Programme on New Trends in PDEs : Theory and Computation, Instructional School on Modern Theory of PDEs at Department of Mathematics, IIT Bombay, Powai, Mumbai-400076, India from 27th May – 23rd June, 2007.
4. National Conference on Biomechanics at Department of Mathematics, IIT Roorkee, Roorkee – 247667, India from March 7-8, 2009.
5. Study Group Meeting on Industrial Problems at Department of Mathematics, IIT Roorkee, Roorkee – 247667, India from March 16-21, 2009.
6. Workshop in the Field of Robotics, Intelligent & Unmanned Systems at NextSapiens, New Delhi from June 15 – 29, 2009.
7. National Meet of Research Scholars in Mathematical Sciences at Department of Mathematics, IIT Roorkee, Roorkee – 247667, India from December 19-23, 2009.
8. International Conference on Image Information Processing (ICIIP -2011) November 3-5, 2011, Jaypee University of Information Technology Waknaghat, Shimla, Himachal Pradesh, India.
9. International Conference on Soft Computing for Problem Solving (SocPros 2011) at the Institution of Engineers (India), Roorkee Local Centre, IIT Roorkee Campus, Roorkee, India from December 20-22, 2011.
10. International Conference on Computer Vision and Image Processing (CVIP 2016) at Department of Computer Science, IIT Roorkee, Roorkee – 247667, India from February 26-28, 2016.
11. International Conference on Frontiers in Industrial and Applied Mathematics (FIAM 2018), NIT Hamirpur, Hamirpur, Himachal Pradesh, India.

12. 4th International Conference on Harmony search, Soft computing and Applications (ICHSA 2018), BML Munjal University, Gurgaon, India.
13. IEEE Sixth International Conference on Multimedia Big Data (BigMM), September 24-26, 2020, New Delhi.
14. 2nd International Conference on Networks and Cryptology (NETCRYPT - 2020), December 4-6, 2020, New Delhi.
15. International Conference on Communication and Computing Systems (ICCCS - 2021), February 5-7, 2021, ICARS, New Delhi.

Awards, Achievements, Memberships and Reviewer:

GATE (Mathematics)	Qualified with All India Rank 201 in 2008.
CSIR-UGC NET (Mathematics)	Qualified with All India Rank 112 in December 2009.
Membership	Institute of Electrical and Electronics Engineers (IEEE), USA.
Reviewer	ISA Transactions (Elsevier), Optical Engineering (SPIE), Applied Mathematical Modelling (Elsevier), IET Image Processing, Mathematics, International Journal of System Assurance Engineering and Management, Symmetry, Journal of Imaging, Syemmetric, Mathematics.
Convenors	1. A workshop on Deep Learning: From Foundations to Cutting- Edge Techniques” from July 15-19, 2024. 2. Faculty Development Program (FDP) on “Effective Pedagogy: Practice and Policy Alignment,” from July 28 to August 01, 2025.
Session Co-Chair	CVIP – 2016, ICHSA – 2018, NETCRYPT- 2020, ICCCS - 2021, ICCSAI – 2023, ICMLDE – 2023, ICICCT – 2024, ICMLDE – 2025
Awards	Best Paper Award (NETCRYPT – 2020), Young Scientist Award (NETCRYPT - 2020) Best Teachers Award (UPES – 2023, UPES - 2024)

Course Taught:

- | | |
|---|--|
| <ul style="list-style-type: none"> • Introduction to Mathematical Logic • Mathematics-I • Mathematics-II • Mathematics-III • Mathematics • Discrete Mathematical Structures • Advanced Mathematics • Applied Numerical Methods • System Modeling and Identification • PDE and System of ODE • Machine Learning | <ul style="list-style-type: none"> • PDE and System of ODE LAB • Discrete Mathematics • Probability and Statistics • Introduction to Data Science • Statistics for Data Science • Digital Image Processing and Machine Vision • Digital Image Processing and Pattern Analysis • Python programming • Applied Statistical Analysis • Document Image Processing • Problem Solving |
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Language Efficiency:

English	Read: Fluent, Write: Fluent, Speak: Fluent
Hindi (Mother Tongue)	Read: Fluent, Write: Fluent, Speak: Fluent

Software Exposure:

Computing Platforms	Windows 7, 10, 11
Programming Languages	C, C++, Python, Matlab, & Mathematica
Tools	Open CV
Software Packages	MS Office, LATEX, SmartDraw